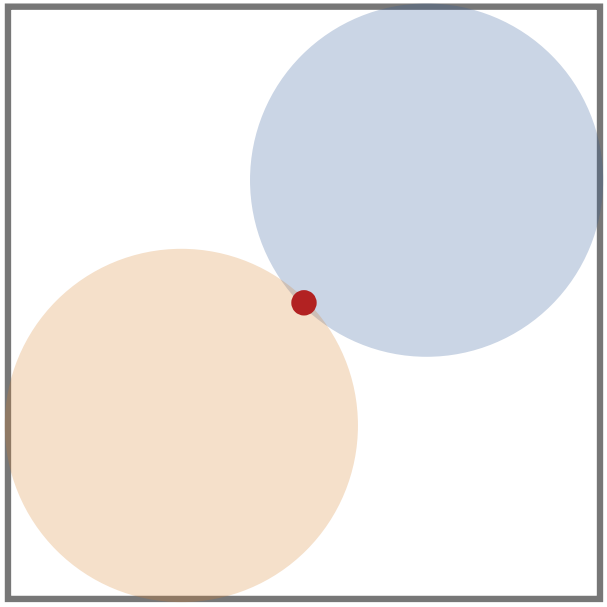
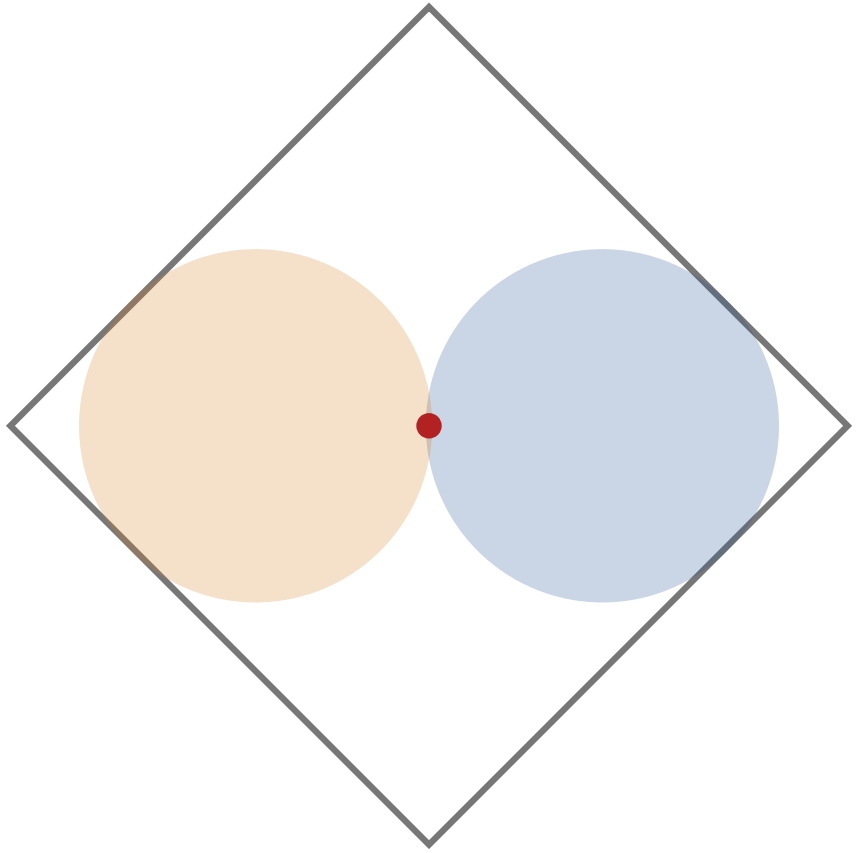
hypercube: $s \geq 2k(1 + 1/\sqrt{d})$



cross-polytope: $s \geq k(1 + \sqrt{d})/\rho$